

## OSWEGO HARBOR, NY, USA

WMO: 997264

Lat: 43.464N

Lon: 76.511W

Elev: 78

StdP: 100.39

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.3      | -13.7 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 24.0                  | -5.6 | 19.7 | -3.2 | 6.5                   | 150  | 0.613 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 6.2                    | 29.1            | N/A  | 27.7 | N/A  | 26.4 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 5.0                  | 170  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 17.5 | 14.7 | 12.4                       | DB  | -18.9              | 32.2 | 2.5                                                | 1.4 | -20.7      | 33.2 | -22.1      | 34.0 | -23.5      | 34.8 | -25.3 | 35.8 |
| (5)               |      |      |                            | WB  | N/A                | N/A  | N/A                                                | N/A | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  | N/A   | N/A  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 9.3    | -4.2  | -3.4  | 0.8   | 7.5   | 13.3  | 18.4  | 21.9  | 21.1  | 17.4  | 11.5  | 5.6   | 0.5  |
|                                                                               | DBStd       | 9.99   | 6.15  | 4.87  | 5.04  | 4.60  | 4.52  | 3.63  | 2.65  | 2.32  | 3.61  | 3.80  | 4.45  | 4.73 |
|                                                                               | HDD10.0     | 1689   | 440   | 376   | 288   | 102   | 17    | 0     | 0     | 0     | 0     | 26    | 144   | 296  |
|                                                                               | HDD18.3     | 3603   | 697   | 609   | 544   | 327   | 169   | 43    | 3     | 4     | 59    | 213   | 382   | 553  |
|                                                                               | CDD10.0     | 1424   | 1     | 1     | 3     | 28    | 118   | 253   | 369   | 343   | 223   | 73    | 11    | 1    |
|                                                                               | CDD18.3     | 296    | 0     | 0     | 0     | 2     | 12    | 46    | 114   | 89    | 32    | 2     | 0     | 0    |
|                                                                               | CDH23.3     | 1399   | 0     | 0     | 0     | 18    | 67    | 229   | 557   | 361   | 158   | 9     | 0     | 0    |
|                                                                               | CDH26.7     | 248    | 0     | 0     | 0     | 3     | 11    | 44    | 97    | 57    | 36    | 1     | 0     | 0    |
| Wind                                                                          | WSAvg       | 5.5    | 6.6   | 6.5   | 6.6   | 6.0   | 4.7   | 4.6   | 4.2   | 4.5   | 4.5   | 5.3   | 5.6   | 6.9  |
| Precipitation                                                                 | PrecAvg     | 1097   | 95    | 78    | 81    | 89    | 88    | 86    | 71    | 91    | 102   | 97    | 118   | 99   |
|                                                                               | PrecMax     | 1388   | 225   | 172   | 142   | 163   | 169   | 181   | 238   | 200   | 208   | 206   | 223   | 166  |
|                                                                               | PrecMin     | 793    | 41    | 17    | 28    | 23    | 26    | 26    | 16    | 32    | 23    | 4     | 46    | 43   |
|                                                                               | PrecStd     | 140    | 38    | 33    | 29    | 38    | 39    | 39    | 42    | 34    | 38    | 44    | 35    | 28   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.6  | 13.4  | 17.7  | 26.1  | 28.2  | 30.0  | 31.1  | 30.1  | 30.6  | 24.4  | 19.3  | 14.6 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | DB     | 10.2  | 8.8   | 14.0  | 21.4  | 25.3  | 27.9  | 29.1  | 28.1  | 27.2  | 21.3  | 16.4  | 11.2 |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | DB     | 6.7   | 5.4   | 11.2  | 17.9  | 22.8  | 26.0  | 27.6  | 26.7  | 25.0  | 19.1  | 14.5  | 8.9  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | DB     | 3.9   | 3.0   | 7.7   | 14.9  | 20.5  | 24.1  | 26.3  | 25.3  | 23.1  | 17.3  | 12.0  | 6.7  |
|                                                                               |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 6.8   | 6.2   | 6.2   | 7.3   | 7.7   | 6.5   | 6.2   | 6.4   | 7.2   | 6.3   | 6.0   | 5.6  |
|                                                                               |             | MCDBR  | 9.6   | 9.6   | 10.5  | 13.6  | 11.7  | 10.1  | 8.3   | 8.8   | 9.3   | 8.9   | 9.3   | 7.9  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                               |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.298  | 0.306 | 0.326 | 0.362 | 0.390 | 0.429 | 0.432 | 0.421 | 0.376 | 0.353 | 0.313 | 0.289 |      |
|                                                                               | taud        | 2.419  | 2.389 | 2.410 | 2.376 | 2.355 | 2.277 | 2.276 | 2.294 | 2.403 | 2.449 | 2.514 | 2.505 |      |
|                                                                               | Ebn at Noon | 835    | 891   | 915   | 903   | 883   | 845   | 838   | 834   | 850   | 825   | 815   | 819   |      |
|                                                                               | Edh at Noon | 75     | 92    | 102   | 115   | 121   | 131   | 130   | 124   | 103   | 87    | 69    | 64    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.20   | 1.98  | 3.26  | 4.47  | 5.52  | 5.84  | 5.95  | 5.29  | 4.17  | 2.46  | 1.50  | 0.93  |      |
|                                                                               | RadStd      | 0.13   | 0.19  | 0.27  | 0.49  | 0.53  | 0.48  | 0.49  | 0.27  | 0.35  | 0.23  | 0.20  | 0.14  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (28 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +72     | N/A     |

Nomenclature: See separate page